

Tensor Network Formulation of the Lagrangian for the Fine-Structure Constant

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Abstract

The fine-structure constant α is a free parameter of quantum electrodynamics. In previous work we showed that α^{-1} can be represented as the on-shell action of a geometric Lagrangian and as the expectation value of a structure operator on a Hilbert space of bounded continued fractions. Here we unify these perspectives through a tensor network formulation, showing that the oscillating circle of radius $R = \pi$ is equivalently described by a circular Matrix Product State (MPS) with bond dimension $D = 45$, where the continued fraction quotients play the role of virtual bonds.

We prove that the commutator between generators at different angular points is identically zero, since $G(\theta) = \mathcal{L}_{\text{geo}}(\theta)I_D + \varepsilon K$ is a sum of an identity matrix (which commutes with everything) and a matrix K that commutes with itself. Consequently, the product of local transfer matrices collapses exactly into the exponential of the sum, with no Baker-Campbell-Hausdorff approximations. The residual $O(1/N^2)$ convergence error is due solely to numerical quadrature of the scalar Lagrangian integral, not to non-commutativity effects.

For $\varepsilon = 0$ and $\mathcal{L}_{\text{curv}} = 0$ (pure bare geometry), the dominant eigenvalue of the contracted transfer matrix yields $\ln(\lambda_{\text{max}}) = 4\pi^3 + \pi^2 + \pi$ with error 10^{-11} at $N = 10^6$ steps, confirming the numerical analysis.

1 Introduction

For more than a century, the fine-structure constant α has been a free parameter of quantum electrodynamics. Its inverse,

$$\alpha^{-1} = 137.035999177(21),$$

is known with extraordinary precision, yet no theoretical derivation from first principles exists.

In previous works [1, 2] we developed three levels of description:

1. A fixed continued fraction representation [14, 1, 7, 3, 1, 3] that reproduces the CODATA 2022 value to 10^{-14} .
2. A quantum expectation model over a Hilbert space of bounded continued fractions (quotients $q_i \leq 45$), with numerical simulation of 1 million collapses at 200-digit precision.

3. A geometric field theory with Lagrangian $\mathcal{L}_{\text{geo}} = 4\dot{d}^2 + (1/R)d^2 + (1/4)\sqrt{R^2 - d^2}$, whose on-shell action for $R = \pi$ yields $4\pi^3 + \pi^2 + \pi \approx \alpha^{-1}$.

In this work we unify these perspectives through the formalism of tensor networks, specifically Matrix Product States (MPS) [3, 5] and Projected Entangled Pair States (PEPS) [4, 6, 7].

2 The Oscillating Circle as a Circular MPS

2.1 Matrix Product States

A Matrix Product State (MPS) for an N -site system is defined as:

$$|\Psi\rangle = \sum_{\{\sigma_i\}} \text{Tr} [A^{(1)}(\sigma_1)A^{(2)}(\sigma_2) \cdots A^{(N)}(\sigma_N)] |\sigma_1, \dots, \sigma_N\rangle,$$

where $A^{(i)}(\sigma_i)$ are $D \times D$ matrices and D is the **bond dimension**. The power of MPS lies in its ability to encode short-range entanglement with a number of parameters scaling linearly with N [8, 9].

2.2 Circle Parameterization

We parameterize the circle of radius $R = \pi$ by the phase angle $\theta \in [0, 2\pi]$. We discretize the cycle into N points $\theta_i = i \cdot \Delta\theta$ with $\Delta\theta = 2\pi/N$. To each point θ_i we associate a tensor $T(\theta_i)$ acting on a virtual space of dimension $D = 45$, where D is the upper bound of the continued fraction quotients.

The continuous generator of the system is:

$$G(\theta) = (\mathcal{L}_{\text{geo}}(\theta) + \mathcal{L}_{\text{curv}}) I_D + \varepsilon K,$$

where:

- $\mathcal{L}_{\text{geo}}(\theta) = 4\dot{d}^2 + (1/R)d^2 + (1/4)\sqrt{R^2 - d^2}$ is the pure geometric Lagrangian density (note the factor 1/4 for the surface term),
- $\mathcal{L}_{\text{curv}} = -\frac{1}{48\pi A(\pi)}$ is the background density due to configuration space curvature,
- I_D is the $D \times D$ identity matrix (the background acts equally on all virtual microstates),
- K is the shift matrix representing the quantum continued fraction operator,
- ε is the entanglement strength (quantum correction).

The local transfer matrix is given by the exact matrix exponential:

$$T(\theta_i) = \exp(G(\theta_i)\Delta\theta).$$

3 Exact Equivalence Proof

3.1 Commutativity Lemma

Lemma 1 (Generator Commutativity). *For any pair of angles $\theta_A, \theta_B \in [0, 2\pi]$, we have:*

$$[G(\theta_A), G(\theta_B)] = 0.$$

Proof. Since $G(\theta) = \mathcal{L}(\theta)I_D + \varepsilon K$, where $\mathcal{L}(\theta) = \mathcal{L}_{\text{geo}}(\theta) + \mathcal{L}_{\text{curv}}$, we compute the commutator:

$$[G(\theta_A), G(\theta_B)] = [\mathcal{L}(\theta_A)I_D, \mathcal{L}(\theta_B)I_D] + [\mathcal{L}(\theta_A)I_D, \varepsilon K] + [\varepsilon K, \mathcal{L}(\theta_B)I_D] + [\varepsilon K, \varepsilon K].$$

The identity matrix I_D commutes with every matrix, and K commutes with itself. Thus all four terms are identically zero. $\square$

3.2 Product Collapse Lemma

Lemma 2 (Product Collapse). *For commuting matrices, the exponential of the sum is exactly the product of the exponentials:*

$$\prod_{i=1}^N \exp(G(\theta_i)\Delta\theta) = \exp\left(\sum_{i=1}^N G(\theta_i)\Delta\theta\right).$$

Proof. By the Baker-Campbell-Hausdorff formula, if $[A, B] = 0$, then $e^A e^B = e^{A+B}$. By induction, the product of N exponentials of pairwise commuting operators is the exponential of the sum. $\square$

3.3 Generator Summation

Lemma 3 (Generator Summation). *Substituting the expression for $G(\theta)$:*

$$\sum_{i=1}^N G(\theta_i)\Delta\theta = \left(\sum_{i=1}^N (\mathcal{L}_{\text{geo}}(\theta_i) + \mathcal{L}_{\text{curv}})\Delta\theta\right) I_D + \varepsilon K \sum_{i=1}^N \Delta\theta.$$

Since $\sum_{i=1}^N \Delta\theta = 2\pi$, the integration of the curvature term yields exactly $2\pi \cdot \left(-\frac{1}{48\pi A(\pi)}\right) = -\frac{1}{24A(\pi)}$. Thus:

$$\sum_{i=1}^N G(\theta_i)\Delta\theta = \left(S_N - \frac{1}{24A(\pi)}\right) I_D + 2\pi\varepsilon K,$$

where $S_N = \sum_{i=1}^N \mathcal{L}_{\text{geo}}(\theta_i)\Delta\theta$ is the approximation of the geometric integral.

3.4 Effective Action Theorem

Theorem 1 (Effective Action). *The dominant eigenvalue of the contracted transfer matrix is:*

$$\lambda_{\max}\left(\prod_{i=1}^N T(\theta_i)\right) = \exp\left(S_N - \frac{1}{24A(\pi)} + 2\pi\varepsilon\mu\right),$$

where $\mu = \lambda_{\max}(K)$ is the dominant eigenvalue of the quantum operator K .

Proof. From the previous lemmas:

$$\prod_{i=1}^N T(\theta_i) = \exp \left(\left(S_N - \frac{1}{24A(\pi)} \right) I_D + 2\pi\varepsilon K \right).$$

For a matrix of the form $cI_D + M$, the eigenvalues are $c + \lambda_i(M)$. Hence:

$$\lambda_{\max} \left(\exp \left(\left(S_N - \frac{1}{24A(\pi)} \right) I_D + 2\pi\varepsilon K \right) \right) = \exp \left(S_N - \frac{1}{24A(\pi)} + 2\pi\varepsilon \cdot \lambda_{\max}(K) \right).$$

□

3.5 Corollaries

Corollary 1 (Pure Geometric Action). *For $\varepsilon = 0$ (no quantum correction) and $\mathcal{L}_{\text{curv}} = 0$ (flat configuration space), in the limit $N \rightarrow \infty$:*

$$\Gamma_{\text{geo}} = \lim_{N \rightarrow \infty} \ln(\lambda_{\max}) = \lim_{N \rightarrow \infty} S_N = \int_0^{2\pi} \mathcal{L}_{\text{geo}}(\theta) d\theta = 4\pi^3 + \pi^2 + \pi.$$

Corollary 2 (Complete Effective Action). *For $\varepsilon \neq 0$:*

$$\Gamma_{\text{eff}} = \int_0^{2\pi} \mathcal{L}_{\text{geo}}(\theta) d\theta - \frac{1}{24A(\pi)} + 2\pi\varepsilon\mu.$$

Setting $\varepsilon = -\frac{\langle \hat{K}^{-1} \rangle}{2\pi A(\pi)^2 \pi^2 \mu}$ yields the exact synthesis of the three contributions:

$$\Gamma_{\text{eff}} = 4\pi^3 + \pi^2 + \pi - \frac{1}{24A(\pi)} - \frac{1}{A(\pi)^2 \pi^2} \langle \hat{K}^{-1} \rangle \approx \alpha^{-1}.$$

4 Numerical Results

We implemented the exact contraction of the circular MPS using the `scipy.linalg.expm` library for matrix exponentiation. The results are summarized in Table 1.

Table 1: Convergence of geometric action with number of steps N

N_{steps}	$S_{\text{geo}} = \ln(\lambda_{\max})$	Error
100	137.0352701653	1.03×10^{-3}
500	137.0362624341	4.13×10^{-5}
1000	137.0362934404	1.03×10^{-5}
2000	137.0363011920	2.58×10^{-6}
5000	137.0363033625	4.13×10^{-7}
10000	137.0363036725	1.03×10^{-7}
1000000	137.0363037759	1.06×10^{-11}

The error scales as $O(1/N^2)$, as expected for a function with cusps at $\theta = \pi/2, 3\pi/2$ due to the $|\cos \theta|$ term in the Lagrangian. The residual convergence error is due solely to numerical quadrature of the scalar integral, not to non-commutativity effects.

5 Connection with Tensor Network Physics

Our formulation finds a natural interpretation in tensor network language:

- The **bond dimension** $D = 45$ corresponds to the upper bound of the continued fraction quotients, observed in all CODATA values from 2006 to 2022.
- The **physical leg** of dimension $d = 1$ indicates that the observable is scalar (the action).
- The **periodicity** of the circle is reflected in the periodic boundary condition of the MPS.
- The **exact exponential** $\exp(G\Delta\theta)$ respects the Lie algebra of the system, ensuring numerical stability and convergence.

The fact that pure bare geometry ($\varepsilon = 0$, $\mathcal{L}_{\text{curv}} = 0$) exactly reproduces the action $4\pi^3 + \pi^2 + \pi$ suggests that the MPS structure captures the essential dynamics of the oscillating circle. Entanglement ($\varepsilon \neq 0$) corresponds to turning on the quantum correction mediated by the operator $\hat{K}$.

6 Future Extensions: 2D PEPS

A natural extension of this work is the construction of a 2D Projected Entangled Pair State (PEPS) [4, 6, 7], where:

- Each angular point θ of the circle becomes a row of a 2D lattice,
- The radial coordinate r (or quotient index) plays the role of the second dimension,
- The observable α^{-1} emerges as the trace of the contracted 2D lattice.

Such a formulation could open the way to:

- Correlation function calculations between quotients,
- Entanglement simulations across different scales of the continued fraction,
- Possible connections with tensor field theory [10] or loop quantum gravity [11].

7 Conclusions

We have proven that the oscillating circle of radius $R = \pi$, described by the geometric Lagrangian $\mathcal{L}_{\text{geo}} = 4\dot{d}^2 + (1/R)d^2 + (1/4)\sqrt{R^2 - d^2}$, can be represented as a **circular Matrix Product State** with bond dimension $D = 45$.

The proof rests on three lemmas:

1. The commutator between generators at different angular points is identically zero.
2. The product of local transfer matrices collapses exactly into the exponential of the sum.

3. The dominant eigenvalue of the contracted matrix yields the desired geometric action.

The residual $O(1/N^2)$ convergence error is due solely to numerical quadrature of the scalar Lagrangian integral, not to non-commutativity effects. With $N = 10^6$ steps, the error is 1.06×10^{-11} , well below experimental precision.

This result unifies the three previous perspectives (continued fraction, quantum expectation, Lagrangian) into a single tensor network formalism, suggesting that the fine-structure constant may be an example of an **emergent observable** in a tensor field theory.

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